

NALLURAJ BABU

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Summary

Applied data science graduate student at the University of Victoria with a computer science degree and a software development internship. Builds analytics end to end: SQL warehouses with data quality tests, statistics that state their limits, dashboards, Python services, and an LLM assistant scored on 40 questions with known answers. AWS Certified Cloud Practitioner.

Technical Skills

- **Languages:** Python, SQL, Java, R, MATLAB, Bash **Web:** HTML, CSS, JavaScript
- **Analytics and Statistics:** Hypothesis Testing, Regression, Forecasting (SARIMA), Cohort Analysis, Power BI, Streamlit, Plotly
- **Data Engineering:** ETL/ELT, dbt, Data Modeling (star schema, SCD Type 2), Data Quality Testing, Airflow, Kafka, Spark
- **Databases and Cloud:** PostgreSQL, MySQL, Microsoft SQL Server, Snowflake, AWS (S3), Docker
- **AI and ML:** LLM Prompting, RAG, LLM Evaluation, Guardrails, Scikit-learn, Sentence-BERT, TensorFlow (coursework), Pandas
- **Engineering Practice:** FastAPI, Unit and Integration Testing, pytest, Git, GitHub Actions CI, Code Review, Agile/Scrum

Projects

ShipSignal: E-Commerce Analytics, Forecasting and LLM Assistant | github.com/NALLURAJ/shipsignal **September 2026**

- Built a PostgreSQL and dbt warehouse over 100k marketplace orders, with staging and mart models and business-rule tests that surfaced 249 payment mismatches and 814 duplicate reviews
- Found that customers whose first order arrived late reordered 1.22% of the time against 1.80% on time (95% CI on the difference -0.92 to -0.24 pp, $p = 0.004$), holding after controlling for state, category and order value
- Forecast daily orders with SARIMA against a seasonal naive baseline (91-day holdout, MAPE 25.1% vs 32.5%)
- Built a natural-language-to-SQL assistant with guardrails (single SELECT, allow-listed tables, row limits, read-only transactions) and scored four prompt versions on 40 questions with known answers, improving 27/40 to 35/40
- Served the results through FastAPI and Streamlit, with 59 pytest tests and CI running ruff, sqlfluff, dbt build and pytest

SkillBridge: NLP Skill Extraction | GitHub | UVic Data Mining, five-person team **Summer 2026**

- Built the skill extractor over 53,906 Canadian job postings, comparing keyword, TF-IDF and Sentence-BERT models over 2,364 skills
- Hand-labelled 100 postings as a gold standard: the training labels matched human judgment at only F1 0.33, which capped accuracy

Airtime Fairness in Wi-Fi 6 (802.11ax) OFDMA | UVic Wireless Networks, two-person team | ns-3, Python **Summer 2026**

- Reproduced the 802.11b performance anomaly in simulation, showed default 802.11ax OFDMA still inherits it (airtime Jain index 0.49), and tested an airtime-deficit scheduler that reached 0.86 and raised throughput 77%

Professional Experience

National Informatics Centre – Ministry of Home Affairs

New Delhi, India

Software Developer Intern

December 2024 – June 2025

- Developed features in Java and Spring Boot for a public-sector platform used by over 10,000 people, in a team of five
- Designed and wrote unit and integration tests that brought the platform to 95% code coverage before release
- Built ETL-based real-time data processing pipelines and maintained database schemas in Microsoft SQL Server
- Performed code reviews and wrote documentation and technical specifications in Agile/Scrum sprints

Education

University of Victoria

Victoria, British Columbia, Canada

M.Eng., Applied Data Science (Graduate Co-op)

Currently Enrolled

Coursework: Optimization for Machine Learning (A+), Algorithms and Data Models (A+), Data Mining, Pattern Recognition

Karunya Institute of Technology and Sciences

Coimbatore, India

Bachelor of Technology, Computer Science and Engineering

Graduated: May 2025

Volunteer Experience

Cook Street Community Centre

Victoria, British Columbia

Volunteer

April 2026 – September 2026

- Helped run community programs and events and worked in the kitchen, about 1 to 3 hours a week

Certifications

- AWS Certified Cloud Practitioner (CLF-C02), 2026 | IBM Data Science Foundation Course, 2024
- IEEE International Conference presentation, 2025: Zero Trust-based intrusion detection and prevention for IoT networks